



American International University-Bangladesh
Faculty of Engineering (FE)
Department of Electrical and Electronic Engineering (EEE)

Course Capstone Project Report

Course Name:	Microprocessor and Embedded Systems	Course Code:	EEE 4103
Semester:	Summer 2025-2026	Section:	
Faculty Name:	Debraj Das		
Capstone Project Title:	IoT-Based Automated Tropical Terrarium Using ESP32		
Project Group #:			

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Assessment Materials and Marks Allocation

CO/CLO Number	CO/CLO Statement	K	P	A	Assessed Program Outcome Indicator	BNQF Indicator	Teaching-Learning Strategy	Assessment Strategy
3	Demonstrate a course project using microcontrollers, sensors, actuators, switches, display devices, etc. that can solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	K8	P1 / P3 / P7		P.d.1.P3	FS.3	Discussion	Project Report (Literature Review)
4	Explain the complex engineering activities of a course project solving a complex engineering problem of the electrical and electronic engineering discipline through an effective presentation.			A1 / A2	P.j.3.A4	SS.2	Discussion	Project Presentation

Assessment Rubrics:

COs	Excellent to Proficient [31-45]	Good [21-30]	Acceptable [11-20]	Unacceptable [1-10]	No Response [0]	Secured Marks
CO3 / P.d.1.P3	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, and more, which can address a complex engineering problem in the electrical and electronic engineering field through appropriate research.	The outcome of the project demonstrates a course project utilizing microcontrollers, sensors, actuators, switches, display devices, etc., and also addresses a complex engineering problem in the electrical and electronic engineering discipline through research.	The outcome of the project demonstrates a course project using microcontrollers, sensors, actuators, switches, display devices, etc. but cannot solve a complex engineering problem properly in the electrical and electronic engineering discipline through appropriate research.	The outcome of the project does not demonstrate a course project using microcontrollers, sensors, actuators, switches, display devices, etc. It also could not solve a complex engineering problem in the electrical and electronic engineering discipline through appropriate research.	No Response at all / copied from others / identical submissions with gross errors / image file printed	
Comments					Total Marks (45)	

IoT-Based Automated Tropical Terrarium Using ESP32

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Abstract—Tropical plants kept indoors need a narrow band of humidity, temperature, soil moisture and light that is difficult to hold by hand, and existing plant controllers are either blind timers or cloud-dependent products. This project designs, builds and tests a self-contained automated terrarium controller around the ESP32 microcontroller. The system senses air temperature, relative humidity and pressure (BME280), light level (BH1750), soil moisture at two points (capacitive probes), tray leakage and reservoir level, and actuates two ultrasonic mist makers, an exhaust fan, a grow light and an alarm buzzer through opto-isolated MOSFET switching stages. A closed-loop rules engine with hysteresis bands, a burst-and-soak watering state machine and a daily watering cap runs entirely on the board; safety interlocks stop misting when the reservoir is empty or a leak is detected. Control is available on a phone through Wi-Fi or the board's own hotspot, and through a Windows console over USB that also installs drivers, flashes firmware and provisions Wi-Fi credentials; a separate tester application exercises every channel by hand. All fourteen automation thresholds are editable at run time and persist in non-volatile storage. Bench testing calibrated both soil probes by two-point measurement (dry 2450 and 2485, wet 1070 ADC counts), established that the MOSFET inputs require a 5 V active-low drive, confirmed flyback protection as mandatory for the fan, and verified end-to-end Wi-Fi provisioning and control at a link strength of about -53 dBm. The electronics cost about BDT 6,200, and a 20-question user survey was designed to capture requirements. The result is a low-cost, internet-independent platform that keeps a tropical micro-climate stable without daily attention.

Keywords—ESP32, Internet of Things, terrarium, closed-loop control, hysteresis, environmental monitoring, home automation

I. INTRODUCTION

A. Background of Study and Motivation

A terrarium is a small, largely enclosed ecosystem in which tropical plants such as ferns, mosses, nerve plants (*Fittonia*) and orchids are grown for decoration and study. Inside the glass the plants create their own water cycle: moisture evaporates from the soil, condenses on the walls and returns,

so the enclosure can stay stable for weeks if the balance is right. These species need 70 to 90 percent relative humidity, a temperature of roughly 22 to 28 degrees Celsius, evenly moist but never waterlogged soil and about twelve hours of light per day. In a Bangladeshi home these conditions are hard to maintain by hand. Dhaka summers reach 33 to 36 degrees Celsius, air-conditioned rooms and the dry season pull indoor humidity down within hours, and the light near a window swings from a few hundred lux on an overcast day to several thousand in direct sun. An anxious owner tends to over-water, which rots the roots; a busy owner forgets, and the plant wilts. Manual care fails silently, because the owner notices the damage only after the plant has declined.

The commercial products that address this problem fall into two groups. Timer-based misting systems act blindly: they mist on a schedule even when the enclosure is already saturated, and they keep misting when the reservoir has run dry, which destroys the ultrasonic disc. Cloud-connected plant monitors, on the other hand, stop working the moment the internet connection or the vendor's service is unavailable, they rarely close the loop on more than one variable, and their thresholds are fixed by the vendor. Neither group offers hardware safety interlocks against an empty reservoir or a leaking tray, and neither can be inspected or repaired by the owner.

The motivation of this work is therefore to build a controller that senses, decides and acts locally on the enclosure itself; that closes the loop on every important variable at once; that can be tuned by a non-expert from a phone; that fails safe; and that remains fully functional with no internet access. Using a microcontroller platform with sensors, actuators, switching devices and a display interface to solve this multi-variable control problem, while balancing cost, safety, power and usability, is precisely the kind of complex engineering activity that the course targets.

B. Project Objectives

The objectives of the project were: 1) to design a fully enclosed, sensor-driven micro-climate control system for a tropical terrarium; 2) to develop ESP32 firmware with a closed-loop rules engine having automatic and manual modes for misting, airflow and lighting; 3) to implement seven sensing channels covering temperature, humidity, pressure, light, dual soil moisture, leakage and water level; 4) to incorporate safety interlocks, namely a reservoir float cut-off, a leak alarm, a daily watering cap and manual time-outs; 5) to provide dual-mode control through a phone dashboard over Wi-Fi or hotspot and a Windows console over USB, with all thresholds adjustable at run time; and 6) to calibrate the soil sensors, characterize the power and switching stages, and analyze the system's behaviour against measured data.

C. Complex Engineering Problem Attributes Addressed

The problem has the attributes of a complex engineering problem as defined by BAETE. It cannot be solved without in-depth knowledge of embedded systems, sensor interfacing, power electronics and control (CP1), and it involves conflicting requirements: high humidity for the plants against condensation on the electronics, frequent misting against reservoir life and disc wear, wireless convenience against offline reliability, and low cost against sensor accuracy (CP2). There is no single obvious solution, since the control laws, thresholds and interlocks had to be formulated and then corrected from measurement, for example when the switching stage proved not to respond to 3.3 V logic (CP3). Finally, the system is a set of interdependent sub-problems, namely sensing, calibration, power, switching, firmware, connectivity, user interface and enclosure, in which a change to one part propagates to the others (CP7).

D. A Brief Outline of the Report

Section II reviews recent journal literature on IoT-based greenhouse, irrigation and plant-monitoring systems and positions this work against it. Section III describes the working principle, the process of work, the components, the power and switching design, the firmware, the software tools and the test setup. Section IV presents the numerical analysis, the measured results, their comparison, the cost analysis and the limitations. Section V concludes the report and lists future endeavours. The as-built wiring diagram, the project timeline and the survey instrument are given in the appendix.

II. LITERATURE REVIEW

A. Related Monitoring and Control Systems

Oguntosin et al. [1] developed an ESP32-based greenhouse monitoring and control system that reads temperature, humidity, a light-dependent resistor and a resistive soil-moisture sensor, and switches a bulb, a fan and a pump whenever a parameter crosses a fixed threshold. Data are stored in a database and shown on a web application. The work demonstrates the basic sense-decide-act loop on the same microcontroller family used here, but its thresholds are compiled into the firmware, control is single-sided with no hysteresis, and there is no protection against an empty water source.

Correa-Quiroz, Toribio-Barrueto and Castro-Vargas [2] reported an ESP32 drip-irrigation and climate-monitoring system for greenhouses using DHT11, capacitive soil-moisture and ultrasonic water-level sensors with an LCD and the Arduino Cloud. Field tests showed a 35 percent reduction in water use, and the system offers both automatic and manual modes, a feature adopted in this project. Remote control, however, depends on a cloud service, so the greenhouse cannot be reached when the internet is down.

Abu Sneineh and Shabaneh [3] built a smart hydroponics monitor on the ESP32 with TDS, pH, water-level and temperature sensors; pumps are actuated automatically when readings deviate from set values, or manually from the Blynk smartphone application. The paper confirms that a single ESP32 can run a multi-variable control loop with a phone interface, but again the interface is cloud-hosted and the actuation decisions carry no time-based safety limits.

Mellit et al. [4] designed a remote greenhouse-monitoring prototype around a NodeMCU that measures air temperature, relative humidity, capacitive soil moisture, light intensity and carbon dioxide, publishes them to a web page, sends GSM alerts and adds a Raspberry Pi camera with a convolutional neural network for disease detection. The breadth of sensing is close to this project, while the two-processor architecture and cellular alerting raise the cost well above what a home terrarium can justify. Simo, Dzitac, Badea and Meianu [7] presented a low-cost IoT greenhouse monitoring device that digitizes environmental and electrical quantities for a farmer's decision making; like most of the surveyed work it is monitoring-oriented and leaves actuation to the user.

B. Sensing and Calibration

Garcia et al. [5] surveyed the sensors and IoT nodes used in smart irrigation and highlighted that low-cost capacitive soil-moisture probes are now accurate enough for irrigation scheduling if they are calibrated in the target medium, and that they outlast resistive probes, whose exposed electrodes corrode in wet soil. This finding motivated the two-point calibration procedure used in Section III and the choice of capacitive probes. For the air, the BME280 combines humidity, temperature and pressure sensing in one I2C device with a specified humidity accuracy of about 3 percent RH [9], which is adequate for a 15 percent hysteresis band; the BH1750 gives illuminance directly in lux over I2C [10], so the light threshold can be set in a physical unit rather than a raw count.

C. Platform, Control Strategy and Connectivity

Hercog, Lerher, Truntic and Tezak [6] documented the design and implementation of ESP32-based IoT devices for teaching, covering the Arduino tool-chain, sensor interfacing over I2C and Wi-Fi connectivity, and their evaluation of the ESP32 as a dual-core, Wi-Fi-capable and inexpensive platform supports its selection as the controller of this project [8]. Across the reviewed systems the control strategy is almost always a single threshold per variable, which causes rapid on-off cycling near the set point and shortens actuator life; a hysteresis band with a minimum run time and a cool-down, as used here, avoids this at no hardware cost. Connectivity in the reviewed work is cloud-first (Arduino Cloud, Blynk, ThingSpeak, GSM), which is convenient for logging but makes the plant depend on a third party.

D. Research Gap

Table I summarizes the comparison. The gap that this project addresses is the combination of multi-variable closed-loop control with hysteresis, hardware safety interlocks, runtime adjustable thresholds, a manual mode with automatic time-outs, and local-first operation that keeps working, and stays controllable from a phone, with no internet or cloud service at all, on hardware costing well under ten thousand taka.

TABLE I. COMPARISON OF RELATED WORK WITH THIS PROJECT

Work	Platform	Sensed variables	Actuation	Offline control	Safety interlocks
[1] 2023	ESP32	T, RH, light, soil	Bulb, fan, pump	No	No
[2] 2025	ESP32	T, RH, UV, soil, level	Drip valve	No (cloud)	Level sensing only
[3] 2023	ESP32	T, pH, TDS, level	Pumps	No (Blynk)	No
[4] 2021	NodeMCU + RPi	T, RH, soil, light, CO2	Irrigation, fan	No	No
[7] 2022	IoT node	Environment, electrical	None	No	No

Work	Platform	Sensed variables	Actuation	Offline control	Safety interlocks
This work	ESP32	T, RH, P, lux, 2x soil, leak, level	2 mist, fan, light, buzzer	Yes (hotspot + USB)	Float, leak, daily cap, time-outs

III. METHODOLOGY AND MODELING

A. Introduction

The system was developed as a bench prototype in stages, each verified with measurements before the next was added: controller and tool-chain bring-up, sensor bring-up and calibration, the power and actuation stage, the firmware rules engine and web interface, the Windows console and provisioning path, and finally a stand-alone manual tester used for the experiments in Section IV. Fig. 1 gives the hardware block diagram, Fig. 2 the software architecture, Fig. 3 the simulated humidity cycle and Fig. 4 the control loop with the watering state machine. The enclosure parts were designed in OpenSCAD for PETG printing and are integrated after the bench phase.

B. Working Principle of the Proposed Project

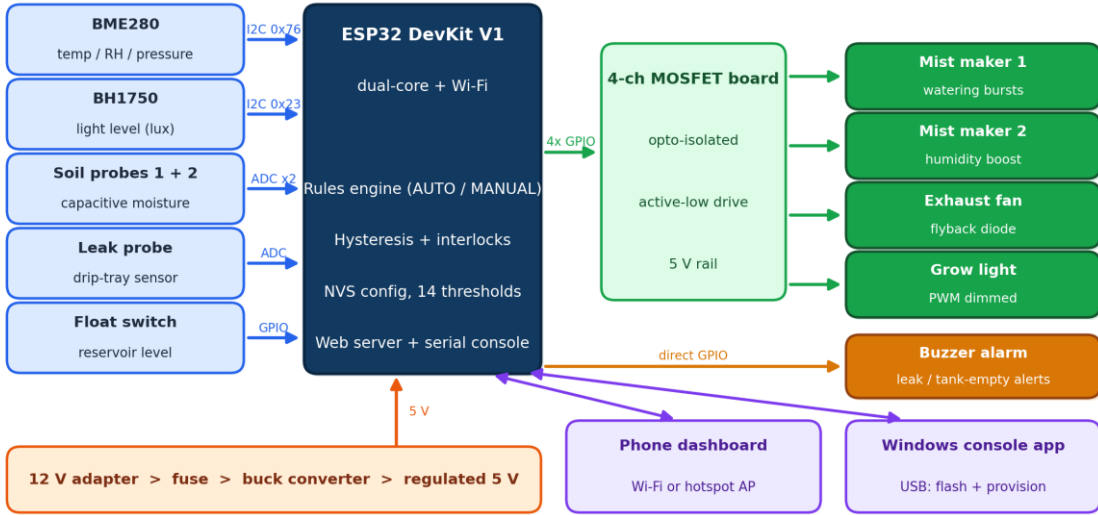


Fig. 1. Hardware block diagram of the automated terrarium system.

Every two seconds the ESP32 reads the BME280 and BH1750 over the I2C bus, samples the two capacitive soil probes and the leak probe on 12-bit analog inputs, and reads the float switch. The readings feed a rules engine that holds the state of five outputs. In automatic mode the engine applies the laws below; in manual mode the outputs follow the user directly while the safety interlocks remain active. Soil moisture is derived from the raw ADC value by the two-point linear calibration

$$M(\%) = 100 (ADC_{dry} - ADC_{raw}) / (ADC_{dry} - ADC_{wet}) \quad (1)$$

where ADC_{dry} and ADC_{wet} are the readings measured with the probe in air and fully submerged in water, and the result

is clamped to 0 to 100 percent. The two probes are averaged for the watering decision so that a single dry pocket does not trigger a burst. Watering follows the burst-and-soak state machine of Fig. 4: when the average moisture falls below the dry threshold (default 35 percent) and the reservoir is not empty, mist maker 1 runs for one burst (default 90 s), the controller then waits a soak period (default 20 min) for the water to spread through the substrate, and re-judges the soil. A hard cap (default 30 min of misting per day) bounds the water that can be delivered even if a probe fails in the dry direction; the daily total resets at midnight when a network clock is available and every 24 h of uptime otherwise.

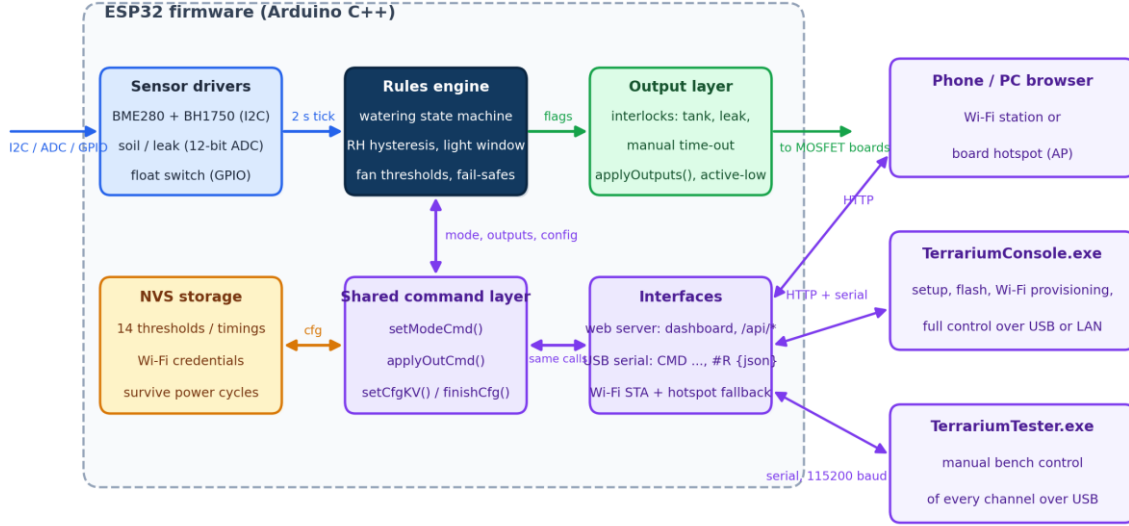


Fig. 2. Firmware and software architecture: one shared command layer serves the web dashboard, the USB console and the tester.

Humidity uses a hysteresis band: mist maker 2 turns on when RH falls below RH_{low} (default 75 percent) and off when RH rises above RH_{high} (default 90 percent), with a maximum burst length of 5 min and a cool-down of 3 min between bursts. Fig. 3 shows the simulated behaviour of this law. The band prevents the rapid on-off chatter that a single threshold would produce, and the cool-down lets the sensor catch up with the mist it has just released. Three consecutive failed humidity readings switch the mist off rather than on, because a dropped I2C read is not evidence that the air is dry.

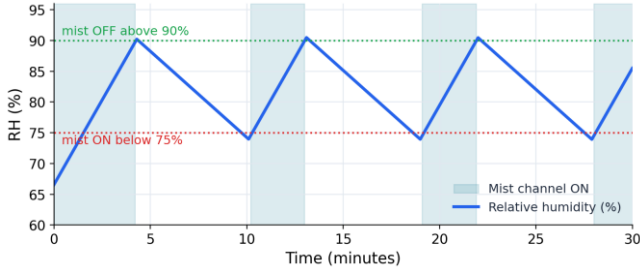


Fig. 3. Simulated humidity hysteresis cycle with the default 75 / 90 percent band.

The grow light follows a photoperiod window (default 07:00 to 19:00, clock from NTP when available) gated by the measured light level: it switches on below 800 lx and off above 2000 lx inside the window and is always off outside it; without a clock the lux gate acts alone. The exhaust fan turns on above 32 degrees Celsius and off below 29. Three interlocks override everything: an empty reservoir disables both mist channels, a wet leak probe raises the buzzer alarm, and any manual mist command is cancelled automatically after ten minutes.

C. Process of Work (Experimental and Simulation Processes)

Work started on 11 August 2026 and followed the timeline in the appendix. Stage 1 set up the tool-chain around the CP210x USB driver and the Arduino ESP32 core (version 3.3.11), driven from the command line so that every build and every serial trace could be captured; a blink test confirmed flashing on COM3. Stage 2 brought up the sensors with an I2C bus scan and a line-health sketch that toggles SDA and SCL and reads them back; both lines read low until the sensor headers were pressed, which located unsoldered header pins as the first fault. The BME280 was found only after its chip-select and address pins were tied to fixed levels before power-up, because the device latches its interface mode at its own power-on. The soil probes were then calibrated by the two-point method in air and in water. Stage 3 added the power chain and the MOSFET board; the drive arrangement was established by measurement (Section IV-B) and the fan test exposed the need for flyback diodes. Stage 4 wrote the rules engine, the on-board dashboard and the USB command protocol; a firmware panic that appeared as two-thirds packet loss on the dashboard was traced with the ESP32 back-trace decoder to a mismatched printf format and fixed. Stage 5 produced the Windows console, which automates driver and tool-chaining installation, firmware flashing and Wi-Fi provisioning over the cable, and Stage 6 the stand-alone tester used for the manual channel tests, at which point a live Fittonia plant was placed in the prototype enclosure (Fig. 5(b)).

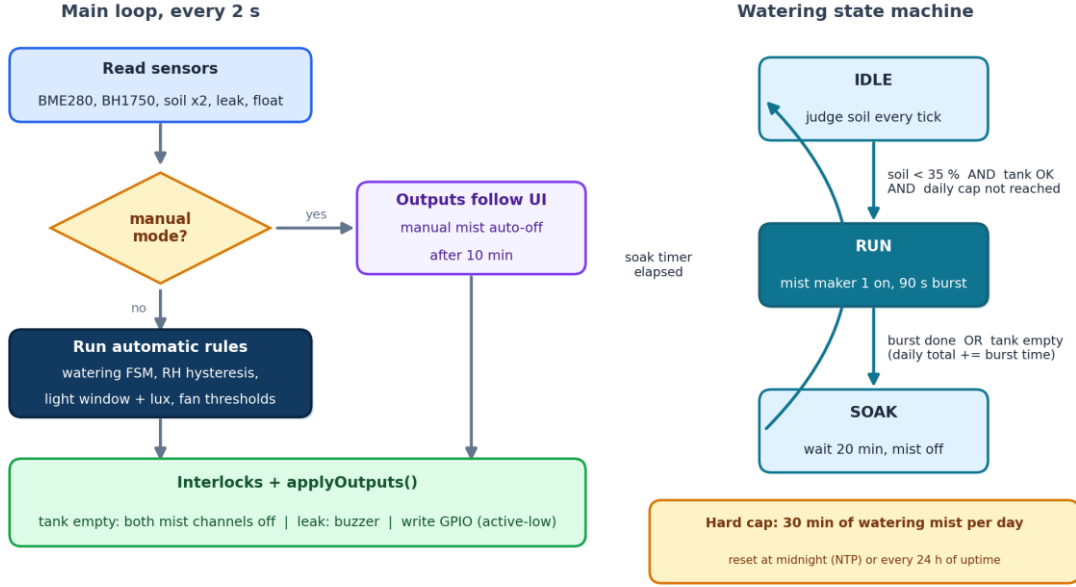


Fig. 4. Control loop executed every 2 s (left) and the watering state machine (right).

The simulation process consisted of modelling the humidity hysteresis law as a discrete-time loop with a constant mist-on rise rate and a constant leak-down rate to predict the cycle period and duty (Fig. 4), of a watering-budget calculation for the state machine, and of a power-budget calculation for the 12 V supply (Section IV-A). In parallel, a 20-question user survey titled Automated Terrarium System Survey was designed on Google Forms to capture requirements (appendix, Table XII): respondent background, indoor plant-care habits and failures, the usefulness of each automation feature on a 1 to 5 scale, mobile-application and alert preferences, safety trust and an acceptable price range for Bangladesh. The survey shaped three design choices: both automatic and manual modes, a prominent empty-tank and leak warning, and a price target below BDT 12,000.

D. Description of the Components

Table II lists the main components with their role and Table III the pin assignment. The ESP32 DevKit V1 provides two 240 MHz cores, 4 MB of flash, Wi-Fi and a CP2102 USB-serial bridge that doubles as the console link; the firmware occupies about 80 percent of the default application partition. The GY-BME280 module measures temperature, relative humidity and pressure over I2C at address 0x76; the BH1750FVI measures illuminance over I2C at address 0x23 with its address pin grounded. The capacitive soil probes and the leak probe are read on 12-bit ADC inputs of the first ADC unit, which are input-only pins chosen so that the Wi-Fi radio, which disables the second ADC unit, does not disturb them. The float switch is read with the internal pull-up and its empty state is configurable. Two 4-channel opto-isolated MOSFET boards switch the loads: board A on the 5 V rail for the two ultrasonic mist makers (DC 5 V, 1 A driver modules that start automatically when powered) and the buzzer, board B on 12 V for the fans and the LED grow light. A 12 V, 5 A adapter feeds a fused LM2596S buck converter that produces the 5 V rail, and a passive buzzer driven at 2.7 kHz provides the alarm. Schottky diodes (1N5819) are fitted across each fan as flyback protection.

TABLE II. MAIN COMPONENTS AND THEIR FUNCTION

Component	Interface / rating	Function
ESP32 DevKit V1	CP2102 USB, Wi-Fi	Controller, web server, USB console
GY-BME280	I2C 0x76, 3.3 V	Air temperature, RH, pressure
BH1750FVI	I2C 0x23, 3.3 V	Illuminance (lux)
Capacitive soil probe x2	ADC, 3.3 V	Soil moisture at two points
Leak probe	ADC, 3.3 V	Drip-tray leak detection
Float switch	GPIO, pull-up	Reservoir empty interlock
MOSFET board x2	Opto-isolated, 60N03	5 V and 12 V load switching
Mist maker x2	DC 5 V, 1 A driver	Watering burst, humidity
Fan WDM4010S12	12 V, 0.08 A	Exhaust / circulation
LED strip 5050	12 V, IP65	Grow light, PWM dimmed
Passive buzzer	2.7 kHz tone	Alarm
12 V 5 A adapter + LM2596S	12 V in, 5 V out	Power supply

TABLE III. ESP32 PIN ASSIGNMENT

Signal	GPIO	Direction	Note
I2C SDA / SCL	21 / 22	bus	BME280, BH1750
Soil probe 1 / 2	34 / 35	ADC in	input-only pins, ADC1
Leak probe	32	ADC in	wet above 1500 counts
Float switch	27	in, pull-up	tank-empty interlock
Mist 1 (water)	16	out, active-low	silk label RX2
Mist 2 (humidity)	25	out, active-low	board A
Exhaust fan	17	out, active-low	silk label TX2
Circulation fan	18	out, active-low	reserved
Grow light	19	PWM, inverted	board B, 12 V
Buzzer	26	out	tone 2.7 kHz
DS18B20 (soil temp.)	4	1-wire	reserved

Signal	GPIO	Direction	Note
Spare channel	23	out	future Peltier

E. Power and Switching Design

The power architecture has three rails. The 12 V rail from the adapter, protected by a 3 A fuse and a rocker switch, feeds the buck converter and the 12 V loads on board B. The 5 V rail from the buck feeds the ESP32 through its VIN pin, the mist driver modules through board A and the opto-coupler inputs of both boards. The 3.3 V rail from the ESP32's on-board regulator feeds the sensors only, because the capacitive probes and the I2C modules are 3.3 V devices. Bulk capacitors of 1000 uF sit at the buck output and at each mist driver so that the inrush of a mist disc does not dip the logic supply.

The MOSFET board is driven active-low. Its opto-coupled inputs need about 5 V to switch, which the 3.3 V logic of the ESP32 cannot supply directly; therefore each channel's PWM terminal is tied to the 5 V rail and its GND terminal is pulled low by the ESP32 pin to turn the channel on, so the microcontroller only ever sinks the small opto-LED current. The firmware inverts every output accordingly, including the PWM duty used to dim the grow light, and writes all outputs to the off state before and after the pins are configured at boot, so that no channel can pulse during a reset. Because the fans are inductive loads, a 1N5819 Schottky diode is placed across each fan with its cathode to the positive lead; the bench test without it reset the controller (Section IV-B). The mist makers are piezo loads and need no diode.

F. Firmware Design

The firmware is written in Arduino C++ for the ESP32 core and is organised as in Fig. 2. A single non-blocking loop samples the sensors every 2 s, runs the rules engine in automatic mode or the manual guards in manual mode, and calls one output function that applies the interlocks and writes the pins. Fourteen configuration values (Table IV) live in non-volatile storage (the ESP32 Preferences API) and can be changed at run time from any client; they survive power cycles and are loaded at boot with compiled defaults as fallback. A shared command layer implements mode changes, output commands and configuration writes exactly once, and both the HTTP handlers of the web server and the USB serial parser call it, so wired and wireless control can never drift apart.

TABLE IV. RUN-TIME CONFIGURABLE PARAMETERS AND DEFAULTS

Parameter	Default	Meaning
soilDry	35 %	water below this average soil moisture
humLo / humHi	75 / 90 %	humidity mist on below / off above
luxOn / luxOff	800 / 2000 lx	grow light on below / off above
lightStart / lightEnd	07 / 19 h	photoperiod window
bright	220 / 255	grow light PWM duty
waterRun	90 s	one watering burst
waterSoak	20 min	wait before re-judging soil
waterCap	30 min/day	hard daily watering limit
humMax	300 s	longest humidity burst

Parameter	Default	Meaning
humCool	180 s	rest between humidity bursts
manMax	10 min	manual mist auto-off

The web server exposes the dashboard page and a small JSON API: status, output, mode, set and wifi endpoints. The USB serial interface accepts the commands of Table V at 115200 baud and answers each with a line prefixed by a marker so that a host can separate replies from the boot log; anything not starting with the command prefix is ignored, so a human typing in a serial monitor does no harm. Wi-Fi credentials saved through either path take precedence over the compiled placeholders; if no network is configured or the join fails, the board raises its own hotspot immediately and keeps retrying the saved network in the background every 90 s, while the clock is re-synchronised from NTP every 5 min until it succeeds. Several defensive details came out of testing: all status strings are built into fixed character buffers rather than temporary String objects, the humidity fail-safe requires three consecutive bad reads, and time comparisons use signed differences so that the 49-day millisecond wrap cannot freeze a cool-down.

TABLE V. USB SERIAL COMMAND PROTOCOL

Command	Reply
CMD STATUS	#R {status json}
CMD OUT <name> <0 1>	#R {status json} or #R {"err":...}
CMD MODE <auto manual>	#R {status json}
CMD SET k=v k=v ...	#R {status json}
CMD WIFI <ssid> <t<pass>	#R {"ok":true}, then restart
CMD WIFI-CLEAR	#R {"ok":true}, then restart
CMD PING	#R {"pong":true}

G. Software Tools

Two Windows applications were built in Python and packaged as single executables so that the system can be set up on any PC without installing anything. TerrariumConsole detects the board over USB or on the local network, installs the CP210x driver and the ESP32 tool-chain if they are missing, compiles and flashes the bundled firmware, provisions Wi-Fi credentials over the cable without re-flashing, shows all sensors live, toggles every output, edits the fourteen parameters and runs an eight-point self-diagnosis. TerrariumTester is a deliberately small bench tool with one large on/off button per channel, an all-off button and the live readings, used for the channel tests in Section IV. The published console carries only placeholder Wi-Fi credentials; real credentials live only in the board's storage and on the user's PC.

H. Test / Experimental Setup

Fig. 5(a) shows the bench setup used for the electrical measurements: the ESP32 and sensors on a breadboard, the MOSFET board and buck converter on the 5 V side, and the mist maker, leak probe and buzzer placed in a plastic tub. Fig. 5(b) shows the working prototype with a live Fittonia plant in a transparent enclosure, a soil probe inserted in the substrate, the air sensor and the fan mounted inside, and the controller tub alongside. A digital multimeter was used for rail voltages

and switching checks, the serial monitor at 115200 baud for the firmware trace, the on-board dashboard on a phone for wireless control, and the tester application for driving each channel by hand while observing the sensors.

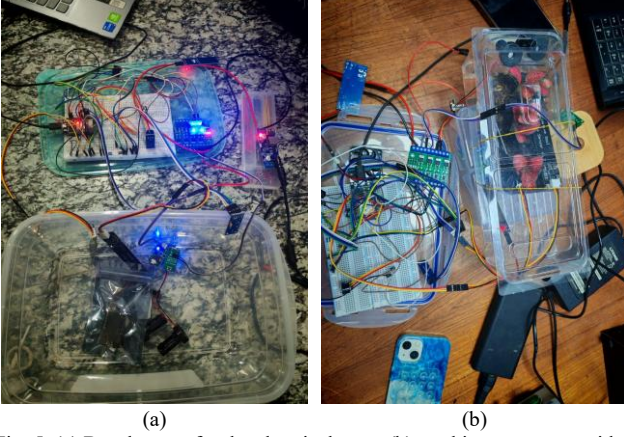


Fig. 5. (a) Bench setup for the electrical tests; (b) working prototype with a live Fittonia plant under automatic control.

IV. RESULTS AND DISCUSSIONS

A. Simulation / Numerical Analysis

Calibration resolution. With the measured end points of probe 1 (2450 counts in air, 1070 in water) the usable span is 1380 counts, so one ADC count corresponds to 0.072 percent moisture; for probe 2 (2485 / 1070) the span is 1415 counts and the resolution 0.071 percent. A reading of 1800 counts on probe 1 maps by (1) to 47.1 percent. The resolution is two orders of magnitude finer than the 35 percent watering threshold, so quantization does not affect the decision.

Watering budget. With a 90 s burst and a 20 min soak the state machine can start at most 2.7 bursts per hour; the 30 min daily cap therefore limits misting to 20 bursts, or about 7 percent of the day, regardless of sensor state. **Humidity cycle.** The simulation in Fig. 3 with a rise of 0.055 percent per second while misting and a fall of 0.028 percent per second otherwise gives a cycle of about 5 min on and 9 min off, a duty of about 35 percent, inside the 5 min burst limit.

Power budget. Table VI sums the rated loads: about 27 W against the 60 W adapter, or 45 percent utilization with every load on at once, and 2.6 A on the 5 V rail against the buck converter's 3 A rating. A Peltier cooler, considered for a future stage, would need its own supply because a TEC1-12706 alone draws up to 60 W.

TABLE VI. POWER BUDGET FROM RATED VALUES

Load	Rail	Current	Power
Mist driver x2	5 V	2 x 1.0 A	10.0 W
ESP32 + sensors	5 V	0.5 A peak	2.5 W
Fan x2	12 V	2 x 0.08 A	1.9 W
LED strip, 1 m	12 V	1.2 A	14.4 W
Total (all on)			28.8 W of 60 W

B. Measured Response / Experimental Results

Table VII gives the soil-probe calibration measured on the bench, and Fig. 6 plots the resulting transfer curves. Both probes gave stable readings within a few counts when held in

air or in water, and responded to wetting within seconds; a disconnected probe floats near 2490 counts, which the firmware reports as dry. Table VIII lists the electrical measurements. The supply chain measured 12.0 V at the adapter and 5.2 V at the buck output under load, and the I2C scan reported the BME280 at 0x76 and the BH1750 at 0x23; the BME280 read room conditions of about 67 percent RH and responded to breath or mist within seconds, and the BH1750 tracked room-light changes.

TABLE VII. MEASURED SOIL-SENSOR CALIBRATION (12-BIT ADC)

Probe	ADC dry (air)	ADC wet (water)	Usable span
Soil 1	2450	1070	1380
Soil 2	2485	1070	1415

TABLE VIII. ELECTRICAL MEASUREMENTS ON THE BENCH

Quantity	Design	Measured
Adapter output	12.0 V	12.0 V
Buck output (5 V rail)	5.0 V	5.2 V
ESP32 3V3 rail	3.3 V	3.2 V
MOSFET input, 3.3 V drive	> 4 V	0.6 to 0.8 V
MOSFET input, 5 V active-low	5 V / 0 V	5.2 V / 0 V
Wi-Fi link strength	> -70 dBm	-53 dBm

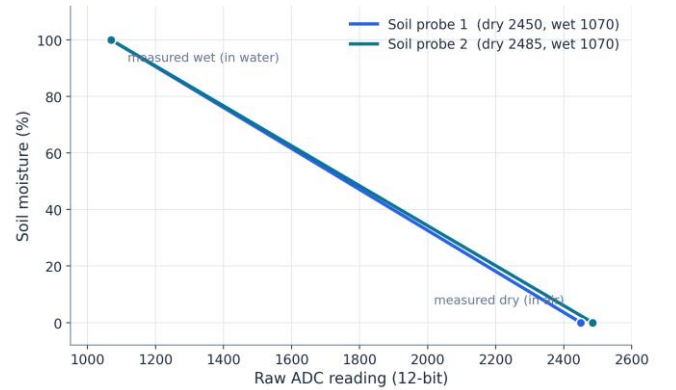


Fig. 6. Measured soil-moisture calibration transfer curves.

Actuation. Driving the MOSFET-board inputs from a 3.3 V pin produced only a dim input LED and no switching; the input terminal measured about 0.6 to 0.8 V across the opto-coupler. With the PWM terminal on the 5 V rail and the GND terminal pulled low by the ESP32 the channel switched cleanly, the input LED lit fully and the mist maker ran from the dashboard, the console and the tester. Switching a fan on the 5 V board without a flyback diode caused the ESP32 to brown out and reset; the same test with the load removed switched the channel LED reliably, confirming the inductive kick as the cause and the diode as the remedy. The buzzer sounded a clean 2.7 kHz tone from the manual button and fell silent on release.

Connectivity. The board joined the home network with a received signal strength of about -53 dBm, and the dashboard was reachable from a phone on the same network. Wi-Fi credentials entered in the console were pushed over USB, after which the board restarted, joined the new network and was rediscovered by the console automatically. With no router present the board raised its own hotspot and the

dashboard was reachable at its fixed address. During development, apparent packet loss of two thirds of requests was traced not to the radio but to a firmware panic on every status request caused by a mismatched format string; after the fix the dashboard served every request. A threshold changed from the console (humCool from 200 s to 180 s) was read back correctly after a power cycle. Table IX summarizes the test matrix.

TABLE IX. TEST MATRIX AND OUTCOMES

Test	Method	Observed	Result
I2C detection	bus scan	0x76, 0x23 found	pass
Line health	SDA/SCL toggle	low until headers soldered	fixed
Soil calibration	air / water	Table VII	pass
Supply rails	multimeter	12.0 V / 5.2 V	pass
Drive at 3.3 V	LED, meter	no switching	fail
Active-low 5 V drive	mist from UI	switches, mist runs	pass
Fan, no diode	toggle	controller reset	fail
Fan channel, no load	toggle	LED switches	pass
Buzzer	UI button	tone on / off	pass
Wi-Fi dashboard	phone	controls, -53 dBm	pass
USB provisioning	console	joined new SSID	pass
Hotspot fallback	no router	AP reachable	pass
Setting persistence	power cycle	value retained	pass
Tester application	each channel	all five toggle	pass

C. Comparison between Numerical and Experimental Results

Table X sets the predicted values against the measurements. The calibration model reproduces the measured end points by construction, and the measured spans confirm that the 12-bit resolution predicted in Section IV-A is available in practice. The hysteresis law behaved as simulated: in automatic mode the humidity channel engaged as soon as the measured RH (67 percent) was below the 75 percent threshold and released only after the upper threshold was crossed or the 5 min burst limit expired, exactly the sequence in Fig. 3; the observed engagement was at first mistaken for a fault until the model explained it. The measured 5.2 V rail is 4 percent above the 5.0 V design value and within the tolerance of the mist drivers and the ESP32 regulator, so the power budget holds. The one prediction that failed, that a 3.3 V pin would drive the opto-coupled inputs, was corrected by measurement and led to the active-low design. A quantitative long-duration comparison of the RH trace against the simulation is planned for the 48-hour soak test described in Section IV-E.

TABLE X. PREDICTED VERSUS MEASURED

Quantity	Predicted	Measured	
Soil resolution, probe 1	0.072 %/count	1380-count span	ok
5 V rail	5.0 V	5.2 V (+4 %)	ok
Humidity engage point	RH < 75 %	engaged at 67 %	ok
Opto drive from 3.3 V	switches	0.6 to 0.8 V, off	no

Quantity	Predicted	Measured	
Fan without diode	switches	controller reset	no
Wi-Fi link	> -70 dBm	-53 dBm	ok

D. Cost Analysis

Table XI lists the cost of the electronics actually used, at the supplier prices paid, together with an estimate for the enclosure materials that are still being sourced. The electronics total about BDT 6,200 and the complete system about BDT 8,200 (roughly USD 70). This falls inside the BDT 8,001 to 12,000 bracket offered as an option in the user survey and is far below imported combinations of a humidistat, an irrigation timer and a lighting controller, none of which provide the interlocks or the phone interface of this design. Because the two switching boards, the second mist maker and the second probe are already included, the marginal cost of a second enclosure on the same controller is only the sensors and actuators inside it.

TABLE XI. COST OF THE BUILT SYSTEM (BDT)

Item	Qty	Unit	Total
ESP32 DevKit V1 (CP2102)	1	650	650
GY-BME280 module	1	420	420
BH1750FVI module	1	235	235
Capacitive soil moisture sensor	2	195	390
Float switch	1	299	299
Leak (water level) sensor	1	49	49
USB mist maker kit with driver	2	380	760
MOSFET F5305S 4-channel board	2	550	1,100
12 V 40 mm fan	2	150	300
12 V 5 A adapter	1	450	450
LM2596S buck converter	1	380	380
Passive buzzer module	1	52	52
1N5819 Schottky diodes (10)	1	30	30
Switch, fuse holders, capacitors, resistors, barrel jack	1	299	299
Veroboard, headers, terminals, jumpers, heat shrink, breadboard	1	795	795
Electronics subtotal			6,209
Enclosure, acrylic, silicone, LED strip, fuses (estimate)	1	1,965	1,965
Total			8,174

E. Limitations in the Project

At the time of writing the system is a verified bench build with a live plant in a temporary transparent enclosure; the printed enclosure and the 12 V board for the light and fans are being integrated, so the results above come from the 5 V stage. The planned 48-hour unattended run, which will provide the RH and soil traces for a quantitative comparison with the simulation and a comparison against a reference hygrometer, is not yet complete, and the survey responses are still being collected. The BH1750 module was unreliable until its header was soldered, and the BME280 requires its mode pins to be fixed before power-up, two assembly details that must be respected in the final build. Humidity is measured at one point, so gradients inside a larger enclosure

are not seen, and the air sensor must be shielded from direct mist to avoid condensation on the element. The ultrasonic mist makers must never run dry, which the float interlock enforces only if the switch is installed as designed. Finally, the controller has no data logging yet, so long-term trends are visible only while a client is watching.

V. CONCLUSION AND FUTURE ENDEAVORS

A complete closed-loop terrarium controller was designed, built and verified end to end on the bench and then demonstrated with a live plant. Seven sensed quantities feed a rules engine on a single ESP32 that drives mist, fan, light and alarm outputs through isolated switching stages, with hysteresis, a timed watering state machine, a daily cap and hardware interlocks providing safe and stable operation. Both soil probes were calibrated, the actuation path was characterized and corrected to an active-low 5 V drive with flyback protection, and wireless and USB control paths including credential provisioning were demonstrated. All fourteen thresholds are tunable at run time, the system operates with no internet connection, and the whole platform costs about BDT 8,200 and is reproducible from the published firmware, tools and wiring documentation. The literature review shows that this combination of multi-variable closed-loop control, safety interlocks and local-first operation is not offered by the reviewed systems.

Future work will integrate the printed PETG enclosure and the 12 V lighting and airflow stage, complete the 48-hour soak test with a reference hygrometer, add on-board data logging with trend charts and push alerts, and evaluate a Peltier cooling stage for the hottest months. Longer-term extensions are camera-based plant-health monitoring, battery backup, a second humidity sensor for gradient detection, and the networking of several enclosures under one console.

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APPENDIX

Fig. 7 reproduces the as-built wiring diagram of the bench build, generated from an interactive page developed for the project in which every wire can be selected and traced. Solid wires are connected and verified; dashed wires are planned. Fig. 8 shows the project timeline, and Table XII lists the survey instrument.

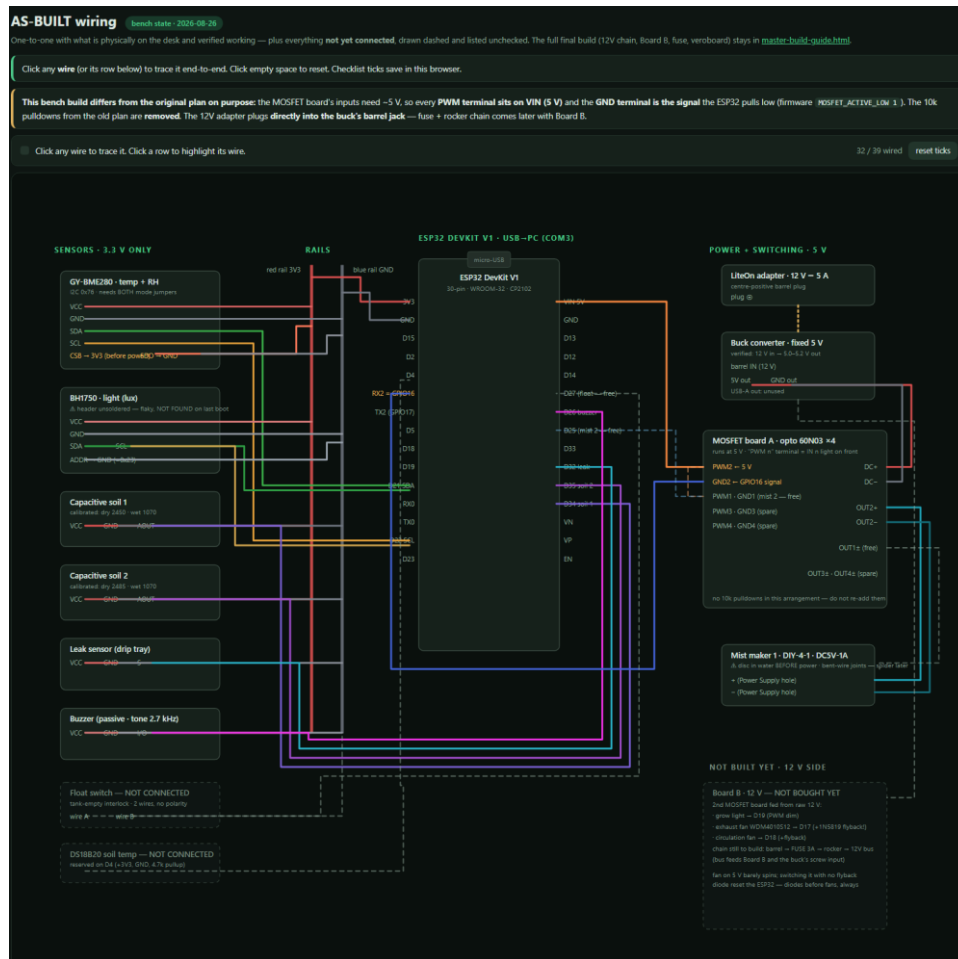


Fig. 7. As-built wiring diagram of the bench build (32 of 39 planned connections live).

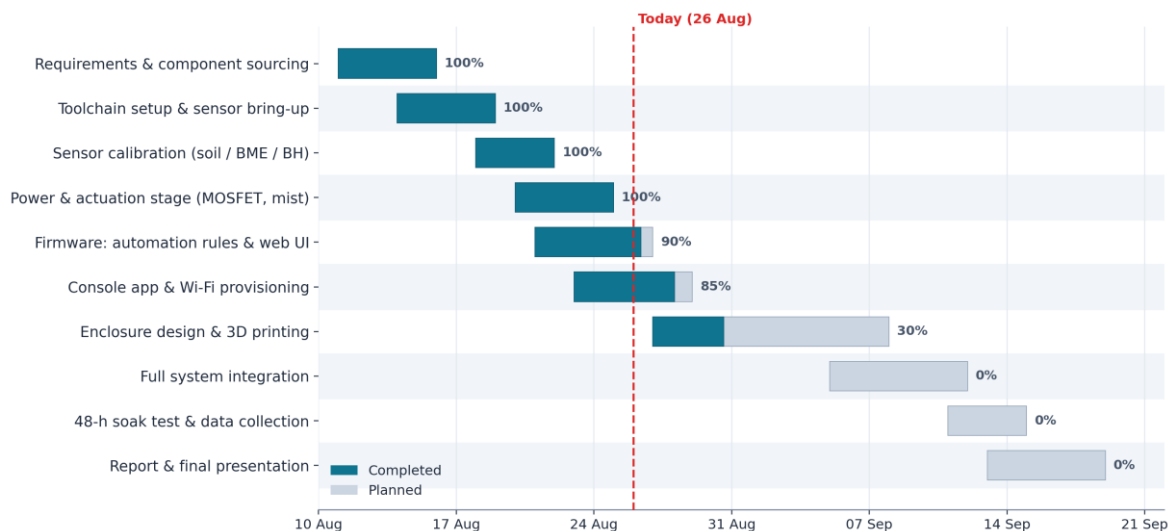


Fig. 8. Project timeline from 11 August 2026 (completed work in dark, planned in light).

TABLE XII. AUTOMATED TERRARIUM SYSTEM SURVEY (GOOGLE FORMS, 20 QUESTIONS)

#	Question	Answer type
1	What is your age group?	choice
2	What is your current occupation?	choice
3	Where do you currently live?	choice
4	Are you familiar with terrariums or enclosed indoor plant ecosystems?	choice
5	Do you currently keep indoor plants at your home, office, or educational institution?	choice
6	How often do you forget to water or care for indoor plants?	scale
7	Have any of your indoor plants become unhealthy or died because of improper watering, lighting, temperature, or humidity?	choice
8	Which environmental factor is the most difficult for you to maintain?	choice
9	How useful would automatic soil-moisture monitoring and irrigation be?	1 to 5
10	How useful would automatic temperature and humidity control be?	1 to 5
11	How useful would automatic grow-light control based on environmental light be?	1 to 5
12	Would you like to monitor the terrarium's temperature, humidity, soil moisture, light level, and reservoir status through a mobile app?	choice
13	Which mobile-app feature would be most useful to you?	choice
14	Would you prefer the system to have both automatic and manual operating modes?	choice
15	Which warning would be most important to you?	choice
16	How concerned would you be about electrical safety when using an automated system containing water, pumps, and electronic components?	1 to 5
17	Would safety features such as low-voltage operation, leakage detection, automatic pump shutdown, and an external electronics box increase your trust in the system?	choice
18	Do you believe an automated terrarium could encourage indoor gardening and environmental awareness among students and urban residents?	Likert
19	Do you believe automatic irrigation could reduce water wastage compared with manual watering?	Likert
20	What price range would you consider reasonable for a complete small automated terrarium system in Bangladesh?	choice

Response charts are appended once collection closes.